Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · Tax ID 0575566002051
www.solaress.org · www.โซลาร์เอสส์.com

QUOTATION

Backup power upgrade: Options A, B and C

Customer	Eco-lodge , Chiang Mai	**Quotation no.**	QO-20261102001
Project	Backup for the whole lodge, water pump and hot water	**Date**	2 November 2026
Existing system	10 kWp solar, 10 kW three-phase hybrid inverter, about 15 kWh lithium battery	**Valid until**	30 days from date

This quotation follows our report Backup power advice (2 Nov 2026). Every option includes Part 1, the backup rewiring. Then choose **one** hot-water option from Part 2: A, B or C.

Part 1: Backup rewiring (included in every option)

#	Item	Description	Qty	Unit	Unit price	Amount
1	**Site survey and load audit**	Check main breaker, list every circuit and its phase, confirm heater and pump ratings	1	Job	–	–
2	**Bypass transfer switch**	Suntree 4P 63 A 400 VAC transfer switch, lets the lodge run from the grid during servicing	1	Pc	–	–
3	**Backup distribution board**	Steel board 400×500, main MCB 4P 63 A, 8 × MCB 2P 32 A, RCCB 4P 32 A Type B, AC SPD 4P, terminals and consumables	1	Set	–	–
4	**Cable**	Yazaki THW 6 mm² (2 rolls black + 1 roll green/yellow) and THW 16 mm² (1 roll) for the inverter backup output and moved circuits	1	Set	–	–
5	**Conduit**	Panasonic IMC 1 in × 10 lengths with waterproof flexible connectors	1	Set	–	–
6	**Water pump connection**	Connect pump circuit to the backup side; soft starter quoted separately only if the survey shows it is needed	1	Job	–	–
7	**Installation labour**	Move lodge circuits and pump to backup, phase balancing, testing	1	Job	–	–
8	**Inverter settings and commissioning**	Battery timetable (40% night reserve / 25% day), grid-charge off, full backup test	1	Job	–	–
9	**Project management**	Transport of materials and team to site	1	Job	–	–
		Subtotal: backup rewiring (all options)				**–**

SOLAR ESS www.solaress.org

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · Tax ID 0575566002051
www.solaress.org · www.โซลาร์เอสส์.com

Part 2, Option A: Heat-pump water heater + storage tank (recommended)

Hot water, the pump and the whole lodge run during a power cut on your existing inverter and battery. Uses about one-third of the electricity of the current heater. One 300 L tank serves about 5–6 showers in a row; if the site visit shows more demand, we will add a second unit.

#	Item	Description	Qty	Unit	Unit price	Amount
10	**Heat-pump water heater, 300 L**	All-in-one heat pump with 300 L insulated tank, about 2.2 kW electrical input, premium brand class (market price, to be confirmed)	1	Set	–	–
11	**Hot-water plumbing**	Connect tank to existing hot-water lines: pipes, valves, mixing valve, insulation	1	Lot	–	–
12	**Heater power circuit**	Suntree MCB 2P 20 A and cable from the backup board	1	Set	–	–
13	**Installation labour**	Set in place, plumbing, commissioning	1	Job	–	–
14	**Remove existing heater**	Disconnect and remove the 11–12 kW heater, make safe	1	Job	–	–
		Subtotal: Option A				**–**

Part 1 + Option A (excl. VAT)	–
VAT 7%	–
Total, Option A	**–**

Part 2, Option B: Electric storage water heater + storage tank

Works on backup with your existing system and costs less to buy. It uses about 3 times more electricity than Option A, so it drains the battery faster during a power cut.

#	Item	Description	Qty	Unit	Unit price	Amount
10	**Electric storage water heater, 300 L**	300 L insulated tank with 3 kW heating element (market price, to be confirmed)	1	Set	–	–
11	**Hot-water plumbing**	Connect tank to existing hot-water lines: pipes, valves, mixing valve, insulation	1	Lot	–	–
12	**Heater power circuit**	Suntree MCB 2P 20 A and cable from the backup board	1	Set	–	–
13	**Installation labour**	Set in place, plumbing, commissioning	1	Job	–	–
14	**Remove existing heater**	Disconnect and remove the 11–12 kW heater, make safe	1	Job	–	–
		Subtotal: Option B				**–**

Part 1 + Option B (excl. VAT)	–
VAT 7%	–
Total, Option B	**–**

SOLAR ESS www.solaress.org

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · Tax ID 0575566002051
www.solaress.org · www.โซลาร์เอสส์.com

Part 2, Option C: Keep the current heater: second inverter + more battery

Doubles inverter power so the existing 11–12 kW heater can run during a power cut. The highest cost, and each shower during a power cut uses a lot of battery.

#	Item	Description	Qty	Unit	Unit price	Amount
10	**Second 10 kW three-phase hybrid inverter**	Deye three-phase LV 10 kW hybrid, run in parallel with your existing inverter (20 kW total, 10 kW per phase). Model to match the existing unit for parallel operation	1	Unit	–	–
11	**Additional lithium battery**	Pylontech Fidus Plus 51.2 V 300 Ah, 16 kWh, IP65, 10-year warranty, matched to your existing battery	1	Unit	–	–
12	**Parallel kit and AC protection**	Parallel communication cables, AC MCB 4P 63 A × 2, terminals	1	Set	–	–
13	**Battery DC cables and DC breaker**	Battery cables 50 mm² and battery DC breaker for the second battery	1	Set	–	–
14	**Installation and parallel commissioning**	Mounting, wiring, parallel setup and full backup test	1	Job	–	–
15	**Project management**	Transport of inverter and battery to site	1	Job	–	–
		Subtotal: Option C				**–**

Part 1 + Option C (excl. VAT)	–
VAT 7%	–
Total, Option C	**–**

Summary

Option	What you get	Before VAT	VAT 7%	Total (THB)
A	Heat-pump water heater + storage tank (recommended)	–	–	**–**
B	Electric storage water heater + storage tank	–	–	**–**
C	Keep the current heater: second inverter + more battery	–	–	**–**

Terms and conditions

- This is an estimate based on current equipment prices. The final price is set after the site survey (heater demand, pump size, main breaker, cable runs) and once you have chosen your option.
- Prices are turnkey: equipment, transport, installation, testing and commissioning.
- **Payment:** in instalments as set out in the contract. Installation labour is subject to 3% withholding tax. Solar ESS issues full e-Tax Invoices.
- **Completion:** within 45 working days from signing and receipt of the first payment (rain days and days off not counted). Actual work on site takes about 3–5 days.
- **Warranty:** 5 years on workmanship by Solar ESS; equipment as per manufacturer (inverter and battery 10 years).
- **Scope:** the solar, battery and backup supply only; faults in other parts of the building's electrical system or in appliances are not covered. Work is protected by CAR insurance; qualifications and certificates are furnished upon request or upon completion of contract.

Customer acceptance — option chosen: A / B / C Signature ..... Name ..... Date	Quoted by: Solar ESS Co. Ltd. Solar ESS Working Group

SOLAR ESS www.solaress.org